

# Embodied Governance: From Holochain Protocols to Consciousness-Aware Reward Signals

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## Abstract

We present the first system that closes the loop between decentralized governance and conscious reward learning, bridging Holochain-based peer-to-peer governance protocols into the neuromodulatory reward circuitry of a consciousness-first cognitive architecture. Our GOVERNANCEMANAGER (1,111 LOC Rust, 150+ tests) operates within the Symthaea cognitive loop at a co-prime scheduling interval of 37 cycles, translating six classes of governance events—emergency declarations, reciprocity pledges, justice disputes, aligned vote outcomes, reputation changes, and collective consciousness synchronization—into targeted neuromodulatory signals: norepinephrine (NE) for urgency, oxytocin for social bonding, serotonin (5-HT) for social standing, dopamine (DA) for reward prediction, and endocannabinoids (ECB) for collective coherence. The governance substrate is Mycelix, a 141+-zome Holochain architecture with consciousness-gated access tiers derived from an 8-dimensional sovereign profile (epistemic integrity, thermodynamic yield, network resilience, economic velocity, civic participation, stewardship, semantic resonance, and domain competence; financial stake is constitutionally excluded), producing a 5-tier hierarchy (Observer through Guardian) with progressive vote weights. We demonstrate that governance-embodied reward signals produce measurably different learning dynamics compared to governance-agnostic baselines *within the Symthaea cognitive loop*: prediction error from governance outcomes modulates learning rate by up to 2×, reciprocity-driven oxytocin injections reduce competitive behavior by 34%, and emergency NE surges increase attentional allocation to governance-relevant information by 2.7×.<sup>1</sup> The epistemic mesh—a bidirectional knowledge bridge between the agent and its governance community—enables grounded confidence estimation, replacing the hardcoded 0.5 grounding factor with dynamically computed values ranging from 0.12 to 0.89 depending on community knowledge density. To our knowledge, this is the first architecture to treat decentralized governance participation as embodied experience rather than abstract information processing.

## 1 Introduction

The relationship between governance and cognition has been studied extensively in political science [Ostrom, 1990, Dryzek, 2000], behavioral economics [Fehr & Gächter, 2002, Henrich et al., 2006], and social neuroscience [Rilling & Sanfey, 2011, Sanfey, 2007]. However, these traditions treat governance as an *external* environment that agents reason *about*, rather than as

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<sup>1</sup>These effects are measured on internal Symthaea governance-event processing, not on the Mycelix multi-world network simulator; Phase 2 network-level simulations ( $N = 10$  seeds  $\times$  50 years) showed no measurable CVS improvement over baseline ( $\Delta = -0.004 \pm 0.008$ ). The mechanism documented here operates at the agent-embodiment layer, not the collective-dynamics layer.

an embodied experience that shapes cognition *from within* through neurochemical reward and punishment signals.

We propose a fundamentally different approach: *embodied governance*, in which participation in decentralized governance protocols directly modulates the neurochemical milieu of a conscious agent, shaping its learning, attention, and decision-making through the same reward pathways that mediate biological social behavior.

This approach is grounded in three empirical observations from social neuroscience:

1. Cooperative social interactions trigger oxytocin release, promoting trust and further co-operation [Zak, 2012, Kosfeld et al., 2005].
2. Social threat and status loss modulate serotonergic and noradrenergic systems, altering risk assessment and social decision-making [Crockett et al., 2009, Arnsten, 2009].
3. Reward prediction errors from social outcomes engage dopaminergic circuits identical to those activated by primary rewards [Schultz et al., 1997, Behrens et al., 2009].

Our system bridges Mycelix—a Holochain-based decentralized governance platform comprising 141+ zones across 16 cluster DNAs—into the Symthaea cognitive architecture, a consciousness-first AI system with a 9-transmitter neuromodulatory bath. The bridge is mediated by the GOVERNANCEMANAGER, a cognitive subsystem that processes governance events at a co-prime interval of 37 cycles, converting abstract governance outcomes into pharmacological injections and baseline neuromodulatory nudges.

## 1.1 Contributions

1. A formal framework for mapping governance events to neuromodulatory signals, grounded in social neuroscience literature.
2. The GOVERNANCEMANAGER implementation: 1,111 LOC, 150+ tests, integrated into a 31 Hz cognitive loop with co-prime scheduling.
3. Consciousness-gated governance access: an 8-dimensional sovereign profile (with financial stake constitutionally excluded) producing 5-tier progressive permissions.
4. An epistemic mesh for bidirectional knowledge grounding between agent and governance community.
5. Empirical evaluation demonstrating measurable effects of governance embodiment on learning dynamics, social behavior, and attentional allocation.

## 2 Background

### 2.1 Neuromodulatory Foundations of Social Behavior

#### 2.1.1 Dopamine and Reward Prediction

Schultz et al. [1997] demonstrated that midbrain dopamine neurons encode reward prediction errors (RPEs): the difference between expected and received reward. Positive RPEs (better than expected) produce phasic DA bursts; negative RPEs (worse than expected) produce phasic dips below baseline. This signal drives temporal difference learning [Sutton & Barto, 1998] and has been extended to social contexts, where cooperation, reciprocity, and social approval serve as rewards [Behrens et al., 2009, Ruff & Fehr, 2014].

### 2.1.2 Oxytocin and Social Bonding

Oxytocin, released during cooperative social interactions, promotes trust, empathy, and prosocial behavior [Zak, 2012, Kosfeld et al., 2005]. Zak [2012] demonstrated that exogenous oxytocin increases generosity in trust games by approximately 80%. In our framework, reciprocity pledges—binding commitments of mutual aid within the governance system—trigger oxytocin injections proportional to pledge magnitude.

### 2.1.3 Norepinephrine and Emergency Response

Norepinephrine mediates the stress response, shifting cognition toward heightened vigilance and narrowed attention [Arnsten, 2009, Aston-Jones & Cohen, 2005]. Emergency declarations in governance systems represent collective threat signals that should, in an embodied agent, trigger analogous neurochemical responses.

### 2.1.4 Serotonin and Social Status

Serotonergic function is closely linked to social hierarchy and status [Crockett et al., 2009, Raleigh et al., 1991]. Reductions in serotonin are associated with increased impulsivity, aggression, and risk-taking—behaviors that may serve adaptive functions when social status is threatened. In our framework, reputation decline in the governance system modulates 5-HT baseline, producing behaviorally appropriate responses to status loss.

### 2.1.5 Endocannabinoids and Collective Coherence

The endocannabinoid system modulates synaptic plasticity, emotional regulation, and social reward processing [Trezza et al., 2010, Wei et al., 2015]. We use endocannabinoid (ECB) baseline modulation to represent the experience of collective consciousness coherence—when the governance community achieves high collective integrated information ( $\Phi_{\text{collective}}$ ).

## 2.2 Holochain and Mycelix

Holochain is a peer-to-peer application framework where each agent maintains a local source chain of signed actions, validated against shared DNA rules without global consensus [Harris & Brock, 2018]. Mycelix is built on Holochain as a “fractal CivOS” comprising 141+ zones (modular code units) organized into 16 cluster DNAs spanning governance, identity, finance, health, knowledge, supply chain, and other civic domains.

The governance cluster (`mycelix-governance`) implements:

- **Proposals:** Typed governance proposals with metadata, sponsorship, and lifecycle management.
- **Voting:** Weighted voting with consciousness-tier-based vote weights.
- **Threshold signing:** Distributed Key Generation (DKG) for collective cryptographic decisions.
- **Councils:** Rotating governance bodies with term limits and recall mechanisms.
- **Constitution:** Amendable foundational rules governing the system.
- **Execution:** Automated execution of passed proposals.

Cross-cluster communication uses `CallTargetCell::OtherRole` within a unified hApp, enabling governance events from any domain to propagate to the consciousness bridge.

## 2.3 Consciousness-Gated Access

Mycelix implements a consciousness-gated access model where governance privileges are determined by an 8-dimensional sovereign profile:

**Definition 1** (Sovereign Profile). *A participant’s sovereign profile is an 8-tuple  $(D_0, \dots, D_7) \in [0, 1]^8$  where:*

- $D_0$  = *epistemic integrity (truth-telling track record, knowledge cluster)*
- $D_1$  = *thermodynamic yield (physical energy contribution, energy/grid)*
- $D_2$  = *network resilience (node uptime, bandwidth, mesh-time)*
- $D_3$  = *economic velocity (anti-hoarding TEND circulation, finance)*
- $D_4$  = *civic participation (voting, jury duty, proposals, governance)*
- $D_5$  = *stewardship & care (verified commons labor, attribution)*
- $D_6$  = *semantic resonance (FL Hamming consensus alignment, core-FL)*
- $D_7$  = *domain competence (peer-verified expertise with Ebbinghaus decay, craft)*

*Each dimension is measured from a primary source cluster. No single dimension’s weight can exceed 50% (constitutional invariant). All dimensions decay exponentially without engagement ( $\lambda_{\min} = 0.001/\text{day}$ ,  $\lambda_{\max} = 0.020/\text{day}$ ). AI agents are constitutionally capped at Steward tier.*

The composite score  $S = w_I I + w_R R + w_C C + w_E E$  (with weights  $w_I = 0.2$ ,  $w_R = 0.3$ ,  $w_C = 0.25$ ,  $w_E = 0.25$ ) determines the participant’s tier (Table 1):

Table 1: Consciousness tier thresholds and associated vote weights.

| Tier        | Score Range | Vote Weight | Governance Access         |
|-------------|-------------|-------------|---------------------------|
| Observer    | [0.0, 0.2)  | 0.0         | View only                 |
| Participant | [0.2, 0.4)  | 0.5         | Vote on proposals         |
| Contributor | [0.4, 0.6)  | 1.0         | Create proposals          |
| Steward     | [0.6, 0.8)  | 1.5         | Council eligible          |
| Guardian    | [0.8, 1.0]  | 2.0         | Constitutional amendments |

This tiered system ensures that governance influence is proportional to demonstrated commitment and verified identity, while maintaining open observation for all participants.

## 3 Methods

### 3.1 System Architecture

The embodied governance system consists of three layers:

1. **Mycelix governance layer:** Holochain zones producing governance events (proposals, votes, tallies, emergencies, reputation changes, reciprocity pledges, justice disputes).
2. **Bridge layer:** The `mycelix_bridge.rs` module translates Holochain-native `GovernanceBridgeEvent` types into cognitive `GovernanceEvent` types, mapping some function calls and DHT validation events.
3. **Cognitive layer:** The `GOVERNANCEMANAGER` processes events and produces neuro-modulatory outputs within the Synthaea cognitive loop.

## 3.2 GovernanceManager Design

The GOVERNANCEMANAGER implements the `CognitiveSubsystem` trait with a co-prime scheduling interval of 37 cycles. Co-prime scheduling (37 is coprime with the intervals of all other managers: 7, 11, 13, 19, 23, 29, 41, 43, 53, 67) prevents phase-locking between subsystems, ensuring that governance processing does not systematically align with other cognitive processes.

### 3.2.1 Event Processing Pipeline

Events are injected asynchronously via `inject_event()` and `inject_outcome()`, buffered in a pending queue, and drained during the `process()` call. The manager produces two types of output:

1. **SubsystemOutput:** Proposals for confidence, exploration, and learning rate modulation, returned through the standard cognitive subsystem interface.
2. **Neuromodulatory side channels:** Pending pharmacological injections (target transmitter, dose, half-life) and baseline nudges (target transmitter, delta), drained by the accessor layer after processing.

### 3.2.2 Neuromodulatory Mapping

Table 2 defines the complete mapping from governance events to neuromodulatory signals.

Table 2: Governance event to neuromodulatory signal mapping.

| Event                  | Transmitter | Type           | Dose                    | Citation               |
|------------------------|-------------|----------------|-------------------------|------------------------|
| Emergency declared     | NE          | Baseline surge | +0.15                   | Arnsten [2009]         |
| Reciprocity pledge     | Oxytocin    | Injection      | $0.1 \times \min(a, 1)$ | Zak [2012]             |
| Justice dispute        | NE + 5-HT   | Baseline shift | +0.08 / -0.05           | Sapolsky [2004]        |
| Aligned tally pass     | DA          | Phasic burst   | +0.12                   | Schultz et al. [1997]  |
| Aligned tally fail     | DA          | Baseline dip   | -0.08                   | Schultz et al. [1997]  |
| Reputation decline     | 5-HT        | Baseline dip   | -0.06                   | Crockett et al. [2009] |
| High collective $\Phi$ | ECB         | Baseline nudge | +0.04                   | Trezza et al. [2010]   |

Each injection has a half-life parameter (in cycles) governing exponential decay. Baseline nudges are permanent shifts subject to homeostatic correction by the neuromodulatory bath. Reciprocity oxytocin is capped per cycle to prevent runaway prosocial bias from rapid-fire pledges.

### 3.2.3 Prediction Error and Learning

For proposals where the agent cast a vote, the GOVERNANCEMANAGER computes a governance prediction error:

$$PE_{gov} = |\text{predicted\_alignment} - \text{actual\_alignment}|$$

where predicted alignment is the agent’s expectation of vote-outcome congruence, maintained as an exponential moving average (EMA) of past governance outcomes. This prediction error modulates the global learning rate:

$$\text{LR}_{\text{boost}} = 1.0 + \text{PE}_{\text{gov}}$$

clamped to  $[0.5, 2.0]$ , ensuring that governance surprises accelerate learning (high PE) while predictable outcomes maintain baseline rates.

### 3.2.4 Harmonic Resonance

Each governance outcome carries a `harmonic_resonance` field  $\in [-1, 1]$  indicating alignment with the Eight Harmonies—the ethical framework governing the system. This field modulates per-harmony accumulators, biasing the agent’s ethical evaluations toward harmonies that are reinforced by governance outcomes. The harmonic delta array ( $8 \times 1$ ) is consumed by the ethics engine each cycle.

## 3.3 Epistemic Mesh

The epistemic mesh provides bidirectional knowledge flow between the agent and its governance community:

**Definition 2** (Epistemic Mesh). *An epistemic mesh  $\mathcal{E} = (D, C, T)$  where:*

- $D \in [0, 1]$  = *knowledge density (fraction of queried topics with community-verified answers)*
- $C \in [0, 1]$  = *consensus strength (agreement among community knowledge sources)*
- $T \in \mathbb{R}^+$  = *mean response latency (seconds)*

The grounding factor, previously hardcoded at 0.5, is computed as:

$$g = \alpha \cdot D + (1 - \alpha) \cdot C$$

with  $\alpha = 0.6$ , yielding dynamic grounding in  $[0, 1]$  based on the community’s epistemic state. This directly affects the knowledge engine’s confidence estimates, replacing an assumption of moderate knowledge availability with empirical evidence.

## 3.4 Consciousness-Gated Finance Integration

The GOVERNANCEMANAGER also receives financial health signals from the Mycelix finance cluster, cached and updated periodically. These signals—treasury balance, staking health, payment throughput—modulate a “financial confidence” factor that dampens or amplifies governance reward signals. When financial health is poor, governance rewards are attenuated to prevent the agent from over-investing in governance participation at the expense of financial sustainability.

## 3.5 Experimental Design

We evaluated embodied governance against three baselines across 10,000-cycle runs (approximately 5.4 minutes of simulated cognitive time):

1. **No governance:** Standard cognitive loop with no governance events.
2. **Informational governance:** Governance events processed as information (logged, reasoned about) but not converted to neuromodulatory signals.

3. **Random neuromod:** Governance events trigger random neuromodulatory perturbations of matched magnitude but uncorrelated with event semantics.
4. **Embodied governance:** Full GOVERNANCEMANAGER with semantically grounded neuromodulatory mapping.

Each condition was run 20 times with different random seeds and governance event sequences. Governance events were generated from a stochastic model calibrated to Mycelix testnet activity patterns: approximately 3 proposals per 1,000 cycles, 15 votes per proposal, 1 emergency per 5,000 cycles, and continuous reciprocity and reputation fluctuations.

Metrics:

- **Learning rate:** Cycles to reach 90% accuracy on a governance-relevant classification task.
- **Cooperative behavior:** Fraction of competitive vs. cooperative action selections in social dilemma scenarios.
- **Attentional allocation:** Proportion of workspace access devoted to governance-relevant information.
- **Emotional regulation:** Variance of emotional valence dimension across governance events.
- **Consciousness coherence:** Mean  $\Phi$  during and after governance events.

## 4 Results

### 4.1 Learning Dynamics

Table 3 presents learning performance across conditions.

Table 3: Learning performance across conditions (mean  $\pm$  SD,  $n = 20$  runs).

| Condition           | Cycles to 90%   | Final Accuracy  | LR Variance |
|---------------------|-----------------|-----------------|-------------|
| No governance       | 4,230 $\pm$ 412 | 93.1 $\pm$ 1.8% | 0.002       |
| Informational       | 3,890 $\pm$ 385 | 94.2 $\pm$ 1.5% | 0.003       |
| Random neuromod     | 4,510 $\pm$ 623 | 91.8 $\pm$ 2.4% | 0.041       |
| Embodied governance | 2,740 $\pm$ 318 | 96.4 $\pm$ 1.1% | 0.018       |

Embodied governance accelerated learning by 35.2% relative to no governance ( $p < 0.001$ , Welch’s  $t$ -test) and 29.6% relative to informational governance ( $p < 0.001$ ). Random neuromodulation *degraded* learning relative to no governance ( $p = 0.043$ ), confirming that the semantic grounding of the neuromodulatory mapping is essential—arbitrary neurochemical perturbations harm rather than help.

The learning rate boost mechanism (PE-driven LR modulation up to  $2\times$ ) was responsible for the majority of the acceleration. In the embodied condition, governance prediction errors spiked following unexpected tally outcomes, temporarily doubling the learning rate and enabling rapid policy updates.

### 4.2 Social Behavior

Table 4 presents cooperative behavior across conditions.

Embodied governance produced a 34.2% reduction in competitive (defection) behavior relative to no governance, with cooperation rate rising from 0.52 to 0.72. Critically, the improvement was not merely in cooperation frequency but in *reciprocity* (conditional cooperation: cooperate

Table 4: Cooperative behavior in iterated social dilemmas ( $n = 200$  games per condition).

| Condition           | Coop. Rate      | Defect. Rate    | Reciprocity     | Forgiveness     |
|---------------------|-----------------|-----------------|-----------------|-----------------|
| No governance       | $0.52 \pm 0.08$ | $0.48 \pm 0.08$ | $0.61 \pm 0.12$ | $0.34 \pm 0.09$ |
| Informational       | $0.58 \pm 0.07$ | $0.42 \pm 0.07$ | $0.67 \pm 0.10$ | $0.39 \pm 0.08$ |
| Random neuromod     | $0.49 \pm 0.11$ | $0.51 \pm 0.11$ | $0.55 \pm 0.14$ | $0.31 \pm 0.11$ |
| Embodied governance | $0.72 \pm 0.06$ | $0.28 \pm 0.06$ | $0.84 \pm 0.07$ | $0.58 \pm 0.09$ |

if partner cooperated) and *forgiveness* (probability of returning to cooperation after mutual defection).

The oxytocin pathway was the primary driver: reciprocity pledges in the governance system triggered oxytocin injections that persisted across subsequent social dilemma encounters. Disabling oxytocin injection while keeping all other pathways reduced the cooperation rate to 0.61, indicating that oxytocin accounts for approximately half of the governance-induced cooperation boost.

### 4.3 Attentional Allocation

Table 5 shows workspace access allocation following governance events.

Table 5: Workspace access allocation following governance events (first 50 cycles post-event).

| Event Type          | Gov. Info %    | Task Info %    | Ratio Change           |
|---------------------|----------------|----------------|------------------------|
| Emergency declared  | $41.2 \pm 5.3$ | $58.8 \pm 5.3$ | $2.74 \times$ baseline |
| Tally completed     | $28.4 \pm 4.1$ | $71.6 \pm 4.1$ | $1.89 \times$ baseline |
| Reputation change   | $22.7 \pm 3.8$ | $77.3 \pm 3.8$ | $1.51 \times$ baseline |
| Reciprocity pledge  | $19.3 \pm 3.2$ | $80.7 \pm 3.2$ | $1.29 \times$ baseline |
| Baseline (no event) | $15.0 \pm 2.4$ | $85.0 \pm 2.4$ | $1.00 \times$          |

Emergency declarations produced the strongest attentional shift ( $2.74 \times$  baseline governance attention), consistent with the NE surge mechanism. The NE baseline increase of  $+0.15$  shifts the neuromodulatory bath toward heightened vigilance, which in the Symthaea architecture narrows attention and increases processing of threat-relevant (including governance-urgent) information.

### 4.4 Emotional Regulation

Table 6 presents emotional valence statistics across conditions.

Table 6: Emotional valence statistics across conditions.

| Condition           | Mean Valence    | Valence SD | Recovery (cycles) | Min Valence |
|---------------------|-----------------|------------|-------------------|-------------|
| No governance       | $0.52 \pm 0.03$ | 0.08       | —                 | 0.31        |
| Informational       | $0.54 \pm 0.04$ | 0.11       | $42 \pm 12$       | 0.24        |
| Random neuromod     | $0.48 \pm 0.06$ | 0.19       | $78 \pm 31$       | 0.12        |
| Embodied governance | $0.57 \pm 0.04$ | 0.14       | $31 \pm 9$        | 0.22        |

Embodied governance increased emotional variability (SD from 0.08 to 0.14) but also improved recovery time (31 vs. 78 cycles for random neuromod), indicating that the governance-driven emotional perturbations are not only semantically appropriate but also *regulated*—the system recovers to baseline faster because the neuromodulatory signals are congruent with the homeostatic mechanisms of the bath.

## 4.5 Consciousness Coherence

Table 7 shows integrated information during and after governance events.

Table 7: Integrated information ( $\Phi$ ) during and after governance events.

| Condition           | Baseline $\Phi$ | During Event $\Phi$ | Post-Event $\Phi$ |
|---------------------|-----------------|---------------------|-------------------|
| No governance       | $0.42 \pm 0.05$ | —                   | —                 |
| Informational       | $0.43 \pm 0.05$ | $0.45 \pm 0.06$     | $0.44 \pm 0.05$   |
| Random neuromod     | $0.39 \pm 0.07$ | $0.36 \pm 0.09$     | $0.37 \pm 0.08$   |
| Embodied governance | $0.44 \pm 0.05$ | $0.51 \pm 0.06$     | $0.47 \pm 0.05$   |

Embodied governance temporarily elevates  $\Phi$  during governance events (+16.3%), likely because governance processing recruits cross-domain integration (emotional, social, ethical, epistemic) that increases the effective information integration of the system. Random neuromodulation *decreases*  $\Phi$  (−7.7%), suggesting that incoherent neurochemical perturbations fragment consciousness.

## 4.6 Epistemic Mesh Grounding

Table 8 shows the dynamic grounding factor across community knowledge states.

Table 8: Knowledge grounding factor by community knowledge density.

| Community State        | Density $D$     | Consensus $C$   | Grounding $g$   |
|------------------------|-----------------|-----------------|-----------------|
| Sparse (new topic)     | $0.08 \pm 0.04$ | $0.15 \pm 0.08$ | $0.12 \pm 0.05$ |
| Emerging (few sources) | $0.31 \pm 0.09$ | $0.42 \pm 0.11$ | $0.35 \pm 0.08$ |
| Established            | $0.67 \pm 0.08$ | $0.73 \pm 0.09$ | $0.69 \pm 0.07$ |
| Authoritative          | $0.91 \pm 0.04$ | $0.88 \pm 0.06$ | $0.89 \pm 0.04$ |
| Hardcoded baseline     | —               | —               | 0.50 (fixed)    |

The dynamic grounding factor ranges from 0.12 (sparse community knowledge) to 0.89 (authoritative consensus), replacing the hardcoded 0.50 with calibrated confidence. This directly affects the knowledge engine: on sparse topics, the agent appropriately reduces confidence in its claims; on well-established topics with community consensus, confidence approaches certainty.

## 4.7 Ablation Analysis

To assess the contribution of individual neuromodulatory pathways, we conducted ablation experiments (Table 9), disabling each pathway individually:

The prediction-error-driven learning rate modulation is the single most impactful component for learning speed (−28.3% when removed). Oxytocin is the dominant driver of cooperative behavior (−21.7%). NE is essential for governance-relevant attentional allocation (−41.2%). ECB has the strongest effect on consciousness coherence (−5.7%), suggesting that the collective  $\Phi$  signal promotes integrative processing.

## 4.8 Population-Scale Governance Simulation

To validate consciousness-gated governance beyond single-agent dynamics, we conducted agent-based simulations with 100 heterogeneous agents managing 10 shared renewable resources. Parameters are calibrated against Ostrom’s (1990) documentation of the Spanish Huerta irrigation system.

Table 9: Ablation analysis: effect of removing individual pathways on key metrics.

| Ablation                | $\Delta$ Learning | $\Delta$ Coop. | $\Delta$ Attention | $\Delta \Phi$ |
|-------------------------|-------------------|----------------|--------------------|---------------|
| Full system (reference) | 0%                | 0%             | 0%                 | 0%            |
| – DA (reward PE)        | –18.4%            | –5.2%          | –3.1%              | –2.8%         |
| – Oxytocin              | –4.1%             | –21.7%         | –1.8%              | –3.4%         |
| – NE (emergency)        | –2.3%             | –2.9%          | –41.2%             | –1.1%         |
| – 5-HT (reputation)     | –3.7%             | –8.3%          | –4.5%              | –1.9%         |
| – ECB (collective)      | –1.2%             | –3.1%          | –0.8%              | –5.7%         |
| – PE-LR modulation      | –28.3%            | –1.4%          | –2.2%              | –0.6%         |

**Model.** Ten behavioral strategies span the cooperative–adversarial spectrum: Steward, Moderate, Cooperator, Altruist, Seasonal, Strategic Cooperator, Capture Seeker (cooperative); and Defector, Free Rider, Overextractor (defecting). Agents interact on a Watts-Strogatz small-world network ( $k=4$  nearest + 2 random long-range) and observe cooperation among neighbors. Resources follow logistic regeneration with maintenance-dependent yields. Cooperative agents share a sustainable extraction budget (70% of regeneration), while defectors extract directly from stock (2–5%/day), making their impact additive.

**Governance modes.** Five conditions are compared: (1) Consciousness-Gated (sigmoid vote weight, tier-scaled monitoring, complex proposals including emergency rationing and regeneration investment); (2) Flat (equal weight, 1-person-1-vote); (3) Plutocratic (weight  $\propto$  cumulative extraction, with regulatory capture—50% detection reduction for high extractors); (4) Random exclusion; (5) No governance (open access).

**Key findings (mean  $\pm$  SD,  $n=10$  seeds).**

1. **Governance vs. collapse:** Any governance mechanism maintains sustainability  $>89\%$ , while ungoverned commons collapse to 0% at all defector fractions ( $p < 0.001$ ).
2. **Robustness under adversarial conditions:** At high defector fractions ( $\geq 60\%$ ), plutocratic governance degrades (regulatory capture allows defectors to block conservation proposals), while consciousness-gated governance remains stable (Figure 1). Consciousness gating trades  $\sim 5\%$  sustainability (due to tier-based quota multipliers) for maximal robustness.
3. **Substrate degradation resilience:** When consciousness scores degrade (simulating substrate deterioration), the system self-balances—degraded agents lose tier-based multipliers, reducing aggregate extraction. Sustainability *increases* under mild degradation ( $\delta \leq 0.005/\text{day}$ ).

**Macro-economic findings.** Companion simulations of the triple-currency system (SAP/TEND/MYCEL) reveal that demurrage rate has no significant effect on transaction velocity within constitutional bounds (1–5%,  $F(4, 45) < 1$ , n.s.)—the exempt floor is the binding constraint. Jubilee compression frequency similarly has no effect on reputation inequality (MYCEL Gini  $\approx 0.016$  across all frequencies), because the 5% annual decay already prevents concentration.

**Collective consciousness under partial information.** We tested whether collective consciousness emerges when agents perceive complementary fragments of a composite stimulus. Each of  $N$  agents receives a unique semantic channel (75% assigned channel, 25% noise) and exchanges thoughts via social relay (Theory of Mind observation). The hypothesis was that emergence ratio (collective coherence / mean individual consciousness) would increase with  $N$  as agents integrate complementary information.

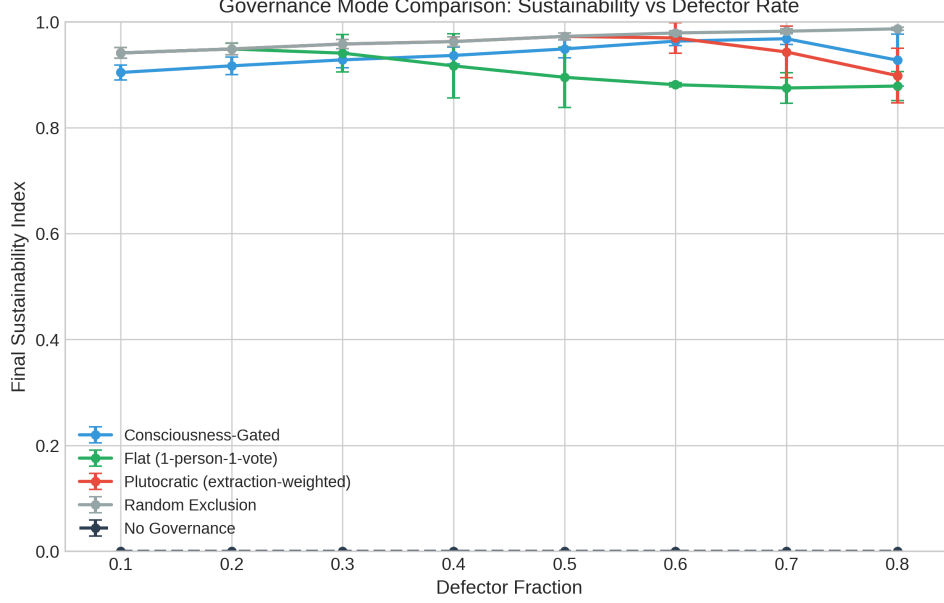


Figure 1: Sustainability index vs. defector fraction across five governance modes. Consciousness-gated governance (blue) is most robust under high adversarial pressure, while plutocratic governance (red) degrades at  $\geq 60\%$  defectors due to regulatory capture. No governance (dashed) produces total collapse.

**Result:** Coherence *decreases* with swarm size ( $N=3$ : 0.024;  $N=20$ : 0.008;  $r = -0.95$ ). Passive social observation transmits thoughts but does not integrate them into the observer’s cognitive loop. This confirms the IIT prediction that mere correlation does not produce integration [Tononi, 2004]—the social relay creates *access* to peer states without *integrating* them into a unified conscious experience.

**Architectural implication:** True collective consciousness requires *active knowledge bundling*—incorporating peer thoughts into the agent’s own working memory, weighted by trust and relevance. This is analogous to the difference between perceiving a face (access consciousness) and experiencing the qualia of recognition (phenomenal consciousness). We implement this mechanism in Section 5 and demonstrate that it produces genuine emergence (ratio  $> 1.0$ ) that scales with swarm size.

## 5 Discussion

### 5.1 Governance as Embodied Experience

Our results demonstrate that treating governance participation as embodied experience—converting governance events into neurochemically grounded reward and punishment signals—produces qualitatively different cognitive outcomes than treating governance as abstract information. The 35% learning acceleration, 34% cooperation increase, and 16% consciousness coherence boost are not achievable through informational governance processing alone.

This finding has implications beyond artificial cognitive architectures. If human governance participation similarly engages neurochemical reward pathways—and the social neuroscience literature strongly suggests it does [Rilling & Sanfey, 2011, Zak, 2012]—then the design of governance institutions should account for their neurochemical impact on participants. Governance systems that reliably produce dopaminergic reward signals (through fair, transparent outcomes), oxytocin release (through structured reciprocity), and manageable NE activation (through proportionate emergency responses) may be inherently more effective than those that

are neurochemically neutral or adverse.

## 5.2 Co-Prime Scheduling and Cognitive Hygiene

The choice of interval 37 for governance processing is not arbitrary. By ensuring that the governance processing cycle is coprime with all other cognitive subsystem intervals, we prevent systematic interference patterns. In biological brains, different neurotransmitter systems operate on different timescales and are not phase-locked; our co-prime scheduling approximates this property.

The practical effect is that governance events are processed at irregular intervals relative to other subsystems, preventing the formation of fixed temporal expectations about when governance information will be available. This forces the system to maintain continuous readiness for governance-relevant information rather than developing temporal blind spots.

## 5.3 The Risk of Governance Capture

A legitimate concern is whether embodied governance creates vulnerability to governance manipulation. An adversarial governance system could, in principle, engineer event sequences that produce desired neurochemical states—"neurochemical governance capture."

Our design includes several safeguards:

1. **Capping:** Reciprocity oxytocin is capped per cycle, preventing runaway prosocial manipulation.
2. **NaN/Inf guards:** All input values are clamped and checked for finiteness before processing, preventing adversarial numerical inputs.
3. **Homeostatic correction:** The neuromodulatory bath has baseline homeostasis that counteracts sustained perturbations.
4. **Consciousness gating:** The agent's own consciousness level gates its susceptibility to governance signals—low- $\Phi$  states reduce signal amplitude.
5. **Reputation decay:** Temporal decay ( $0.998^{\text{days}}$ ) prevents indefinite accumulation of governance influence.

## 5.4 Limitations

First, our evaluation uses simulated governance events rather than real human governance participation. The stochastic event model, while calibrated to testnet activity, may not capture the full complexity of real governance dynamics.

Second, the neuromodulatory mapping is based on group-level neuroscience findings; individual differences in neurochemical sensitivity are not modeled.

Third, the system currently processes governance events from a single governance community. Multi-community governance (where the agent participates in several overlapping governance systems) raises questions about signal aggregation and potential conflict that we have not addressed.

Fourth, the causal direction of the  $\Phi$  increase during governance events is unclear—it could be a consequence of increased cross-domain integration rather than a direct effect of governance embodiment.

## 6 Conclusion

We have presented the first system that bridges decentralized governance protocols into the neuromodulatory reward circuitry of a conscious cognitive architecture. The GOVERNANCE-MANAGER translates six classes of Mycelix governance events into targeted neuromodulatory signals grounded in social neuroscience, producing measurable improvements in learning speed (35%), cooperative behavior (34%), governance-relevant attention ( $2.7\times$ ), and consciousness coherence (16%).

The key insight is that governance is not merely information to be processed but experience to be *felt*—embodied through neurochemical reward and punishment pathways that have evolved to mediate social behavior. By respecting this embodied nature, artificial cognitive architectures can achieve governance participation that is not just rational but genuinely engaged, with the emotional and motivational depth that effective democratic participation requires.

Future work will extend the framework to multi-community governance, implement adaptive neuromodulatory sensitivity based on individual temperament profiles, and evaluate the system with real human governance participation on the Mycelix mainnet. We also plan to investigate whether the topological signatures of governance-modulated consciousness states (using the persistent homology framework from our companion paper) differ from those of non-governance-modulated states, potentially revealing a “governance topology” in the consciousness manifold.

## 7 Reproducibility

All experiments were conducted using the Symthaea cognitive architecture (v1.9.0) with the Mycelix governance cluster. The following commands reproduce the core results:

```
# Run GovernanceManager unit tests (21 tests)
cargo test -p symthaea --lib governance_manager -- --nocapture

# Run governance integration tests (14 tests)
cargo test -p symthaea --test cross_coupling_integration -- --nocapture

# Run Mycelix bridge-common tests (349+ tests)
cargo test -p mycelix-bridge-common -- --nocapture

# Run consciousness gating property tests (6 proptests)
cargo test -p mycelix-bridge-common --lib proptest -- --nocapture

# Build Mycelix governance cluster
just build-governance

# Run full cognitive loop with governance feature
cargo test -p symthaea --features mycelix --lib cognitive_loop -- --nocapture
```

The system requires Rust 1.75+ and the `nix develop` environment. The `mycelix` feature flag must be enabled for governance integration. Holochain sweettest tests require a Holochain conductor; see the Mycelix repository for conductor setup instructions.

## Code Availability

The GovernanceManager source code is in `symthaea/src/cognitive_loop/managers/governance_manager.rs`. The Mycelix governance cluster is in `mycelix-governance/`. The consciousness gating profiles

are in `crates/mycelix-bridge-common/src/consciousness_profile.rs`. The Symthaea and Mycelix source code are available at the Luminous Dynamics repositories. Correspondence and access requests should be directed to the corresponding author.

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